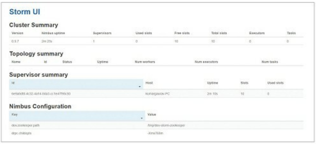

Figure 5: Apache Storm UI with the base configurations, nodes and cluster information

In the MS Windows command prompt, go to the path of *STORM_HOME* and execute the following commands:

```
1.   storm nimbus
2.   storm supervisor
3.   storm ui
```

In any Web browser, execute the URL *http://localhost:8080* to confirm the working of Apache Storm.

Apache Storm is associated with a number of key components and modules, which work together to do high performance computing. These components include Nimbus Node, Supervisor Node, Worker Process, Executor, Task and many others. Table 1 gives a brief description of the key components used in the implementation of Apache Storm.

Table 1

Components	Description
Nimbus	This is the master node in the cluster of Apache Storm. The other nodes are referred to as worker nodes. The master node is the key node used for the distribution of data to the worker nodes and for overall monitoring.
Supervisor	These are the nodes that accept the instructions sent by Nimbus. The supervisor node has multiple processes mapped with the workers.
Worker process	This executes tasks associated with a particular topology. It creates the executors to execute the tasks.
Executor	This is the thread created to execute the process.
Task	This refers to the specific task to be performed.
Zookeeper framework	A service that is used by a group of nodes or simply clusters. Zookeeper assists the supervisor to communicate with Nimbus and maintain the states of communication.
Tuple	This refers to the key data structure in Apache Storm, which supports all formats and data types.
Stream	The sequence of tuples in unordered perspectives.
Spout	This is used to accept the input data from different channels like Twitter streaming, bioinformatics, medical reports, satellite data, etc.
Bolts	These are the logical processing units in Apache Storm.

Extraction and analytics of Twitter streams using Apache Storm

To extract the live data from Twitter, the APIs of Twitter4j are used. These provide the programming interface to connect with Twitter servers. In the Eclipse IDE, the Java code can be programmed for predictive analysis and evaluation of the tweets fetched from real-time streaming channels. As social media mining is one of the key areas of research in order to predict popularity, the code snippets are available at *http://opensourceforu.com/article_source_code/2017/nov/strom.zip* can be used to extract the real-time streaming and evaluation of user sentiments.

Scope for research and development

The extraction of data sets from live satellite channels and cloud delivery points can be implemented using the integrated approach of Apache Storm to make accurate predictions on specific paradigms. As an example, live streaming data that gives the longitude and latitude of a smart gadget can be used to predict the upcoming position of a specific person, using the deep learning based approach. In bioinformatics and medical sciences, the probability of a particular person getting a specific disease can be predicted with the neural network based learning of historical medical records and health parameters using Apache Storm. Besides these, there are many domains in which Big Data analytics can be done for a social cause. These include Aadhaar data sets, banking data sets and rainfall predictions. END

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